

1. Title Page

Title: Statistical Interpretation and Exploratory Data Analysis of the Human Development Index

Student Name: Kshitiz Shrestha

Student ID: 2516667

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3. Introduction

Short Background on the Human Development Index

The Human Development Index (HDI) is a composite indicator to measure the overall level of development of countries using three dimensions: health (life expectancy), education, and standard of living (GNI per capita).

Purpose of the Analysis

- To explore the distribution of HDI across countries.
- To analyze HDI trends from 2020 to 2022.
- To examine relationships between HDI and related indicators.
- To compare development patterns between South Asia and the Middle East.

Coverage and Boundaries of the Study

The analysis uses HDI data from 1990–2022 with emphasis on recent years (2020–2022) and selected regional subsets.

4. Problem-wise Analysis

Problem 1A — Single Year HDI Exploration (2022)

Methodological Framework

The dataset was filtered to 2022, non-country aggregates were removed, country names were standardized, missing HDI values were dropped, and other numeric values were imputed using the median.

Main Findings

- Significant variation exists in HDI across countries.
- A small number of countries fall into the "Very High" HDI category.

169	
country	Switzerland
hdi	0.967

161	
country	Somalia
hdi	0.38

Figure 1: Countries with the Highest and Lowest HDI in 2022

Analysis and Interpretation of Results

The figure highlights the large disparity in human development across countries. Switzerland exhibits the highest HDI value in 2022, reflecting strong performance in health, education, and income. In contrast, Somalia has the lowest HDI, indicating severe challenges in basic human development such as access to healthcare, education, and economic opportunities.

This stark contrast illustrates the uneven distribution of development worldwide and emphasizes that while some countries have achieved very high standards of living, others continue to struggle with fundamental development needs. The difference between the highest and lowest HDI values demonstrates the importance of targeted social, economic, and institutional interventions in low-HDI countries to reduce global inequality.

Problem 1B — HDI Trend Analysis and Visualization (2020–2022)

Methodological Framework

The dataset was filtered to include HDI records for the years 2020, 2021, and 2022. Non-country aggregate entries and records with missing HDI values were removed, and country names were standardized for consistency. Visual and statistical analyses were performed using line charts, bar charts, box plots, and scatter plots to examine trends, regional differences, distributions, and relationships between HDI and income.

Main Findings

Global HDI showed a slight decline in 2021 followed by an increase in 2022, indicating recovery after the COVID-19 pandemic. Europe and Central Asia had the highest average HDI, while Sub-Saharan Africa had the lowest. A strong positive relationship was observed between HDI and GNI per capita, and several countries showed notable improvement between 2020 and 2022.

Visualizations and Figures

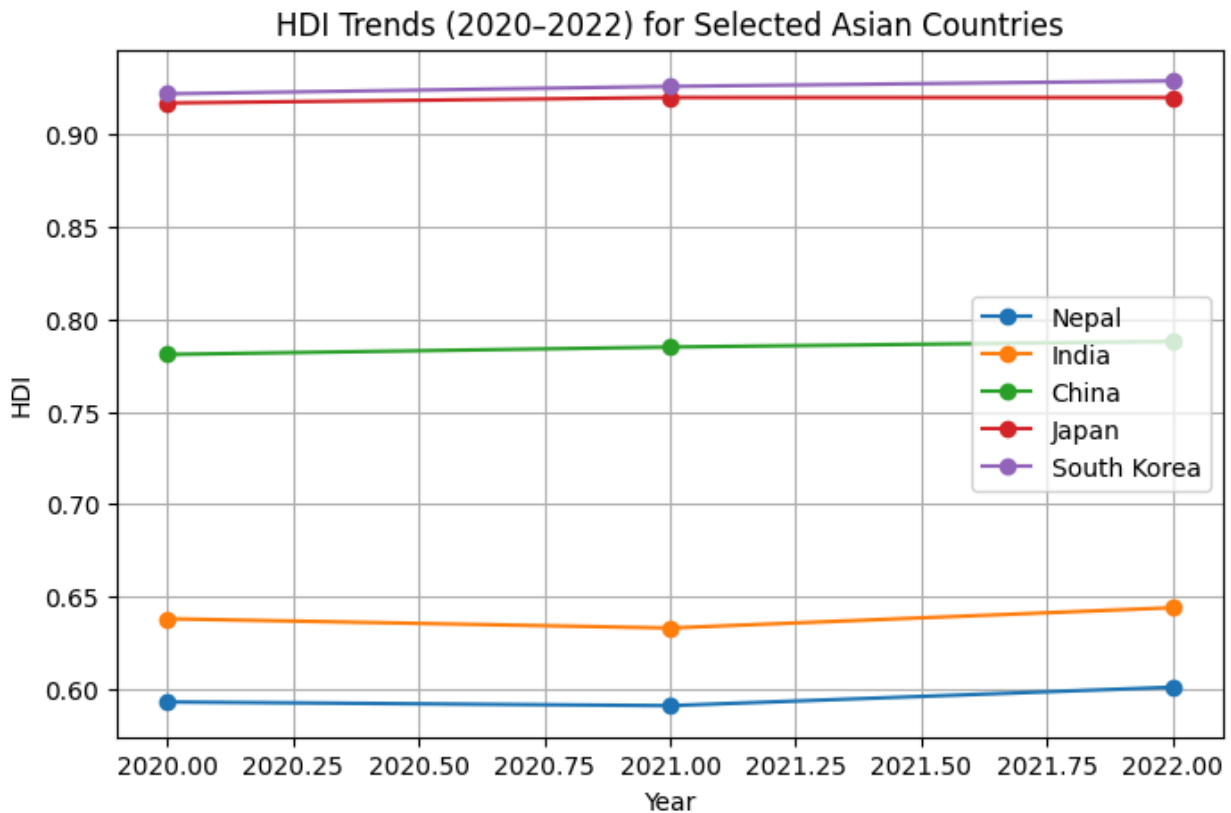


Figure 2: HDI trends from 2020 to 2022 for selected Asian countries (Nepal, India, China, Japan, and South Korea).

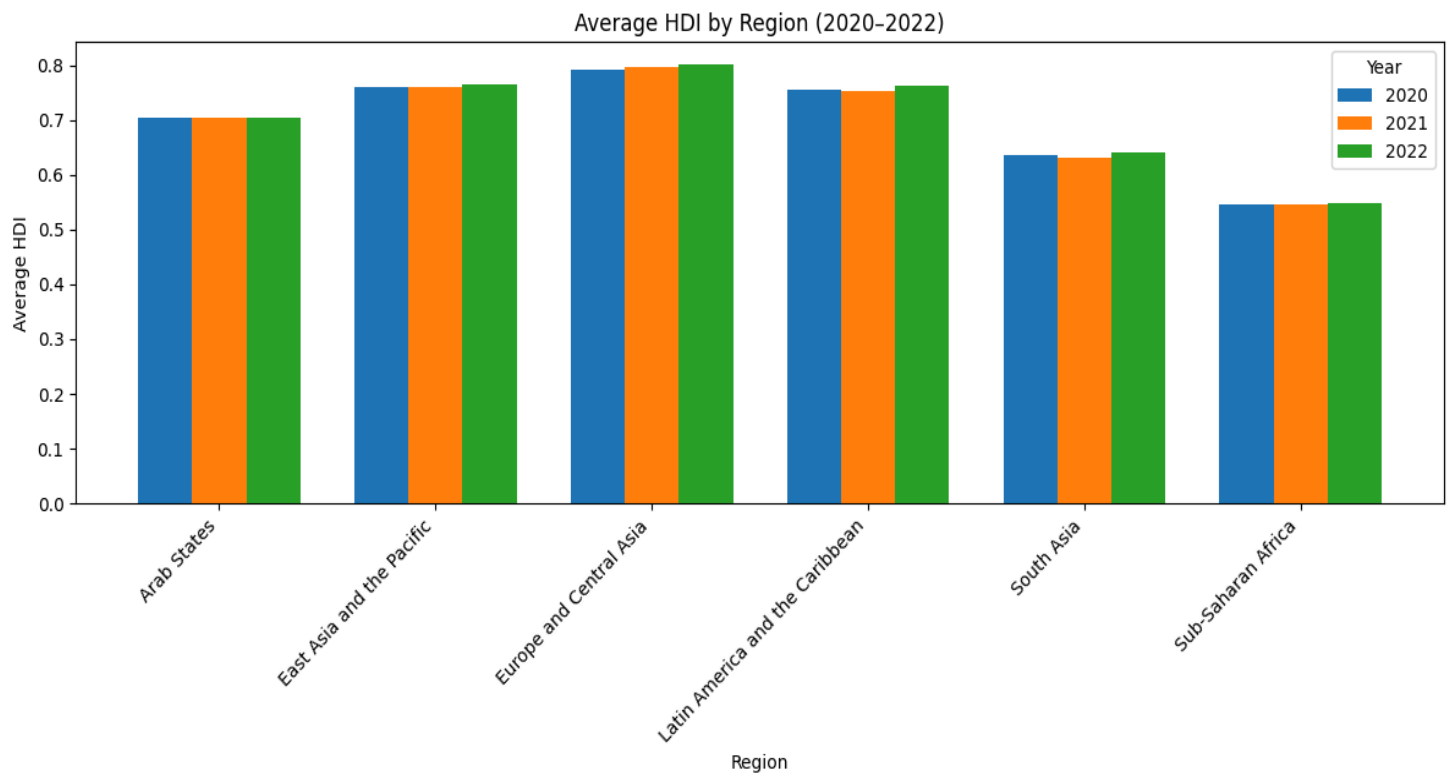


Figure 3: Average HDI by region for the years 2020, 2021, and 2022.

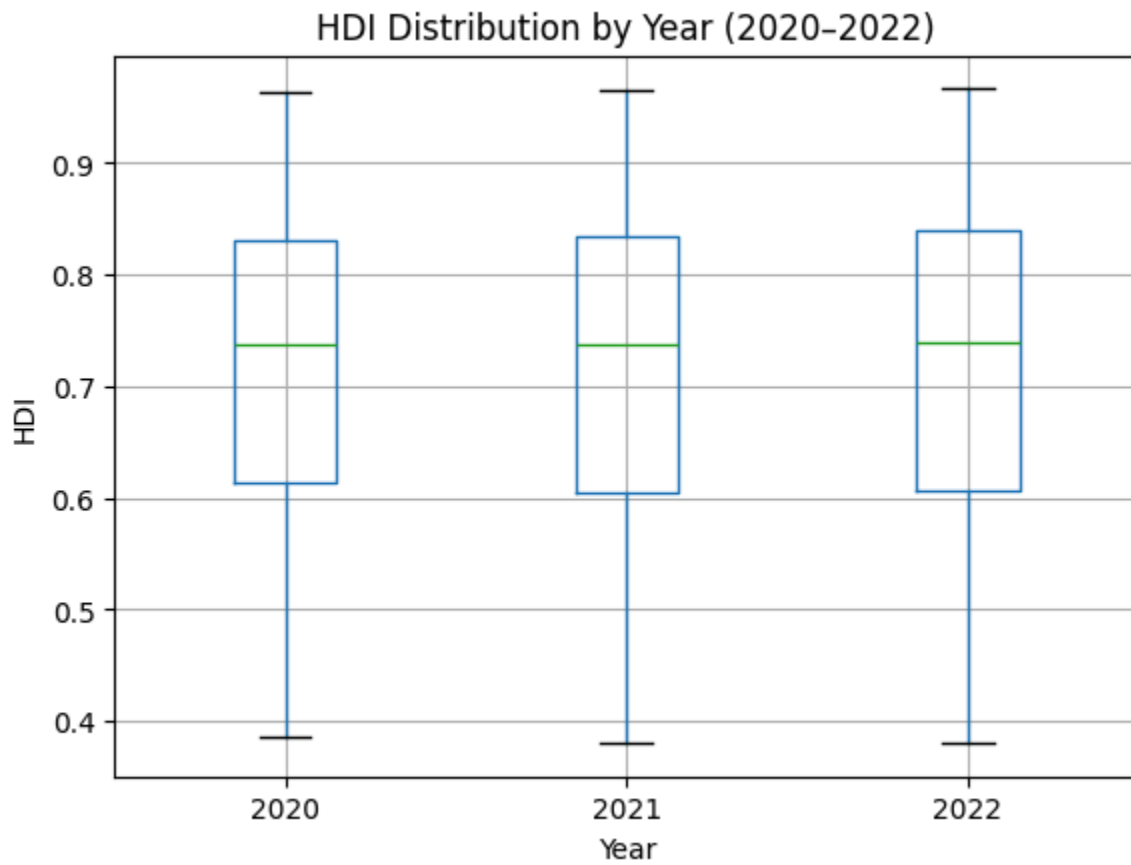


Figure 4: Distribution of HDI values for the years 2020, 2021, and 2022.

year	2020	2021	2022	change_2020_22
country				
Andorra	0.843	0.855	0.884	0.041
Azerbaijan	0.722	0.738	0.760	0.038
Ecuador	0.734	0.746	0.765	0.031
Maldives	0.737	0.753	0.762	0.025
Mexico	0.757	0.757	0.781	0.024
Tajikistan	0.656	0.677	0.679	0.023
San Marino	0.844	0.853	0.867	0.023
Bahamas	0.798	0.799	0.820	0.022
Kuwait	0.826	0.836	0.847	0.021
Türkiye	0.835	0.841	0.855	0.020

Figure 5: Top 10 countries with the greatest increase in HDI from 2020 to 2022.

Analysis and Interpretation of Results

Figure 2 (HDI Trends for Selected Asian Countries) shows that most selected countries experienced small changes in HDI from 2020 to 2022. Nepal and India show slight declines in 2021 followed by recovery in 2022, likely reflecting the impact of the COVID-19 pandemic. Japan and South Korea maintain consistently high HDI values, while China shows a steady increase.

Figure 3 (Average HDI by Region) highlights clear regional disparities, with Europe and Central Asia having the highest average HDI and Sub-Saharan Africa the lowest. South Asia also remains below the global average, indicating ongoing development challenges.

Figure 4 (HDI Distribution by Year) indicates that the overall distribution of HDI values remained relatively stable across the three years, with little change in medians and spread. This suggests that global inequality between countries did not change significantly over this period.

Figure 5 (Countries with the Greatest HDI Improvement) shows that Andorra, Azerbaijan, and Ecuador recorded the largest HDI increases from 2020 to 2022, indicating uneven but positive recovery across countries.

Problem 2 – Advanced HDI Exploration

Methodological Framework

This analysis focuses on South Asian countries: Afghanistan, Bangladesh, Bhutan, India, Maldives, Nepal, Pakistan, and Sri Lanka. A subset of the HDI dataset (1990–2022) was created using these countries.

A composite development score was calculated as:

Composite Score = $0.30 \times \text{Life Expectancy Index} + 0.30 \times \text{GNI per Capita Index}$

Countries were ranked based on their average composite score across years.

Outliers were detected using the Interquartile Range (IQR) method on HDI and GNI per capita values.

Pearson correlation was used to examine relationships between HDI and:

- Life Expectancy Index
- Gender Development Index

A gap analysis was performed by normalizing GNI per capita and computing:

$$\text{GNI_norm} = (\text{GNI} - \text{min}) / (\text{max} - \text{min})$$

$$\text{GNI-HDI Gap} = \text{GNI_norm} - \text{HDI}$$

Main Findings

- Maldives, Sri Lanka, and Bhutan rank highest in the composite development score.
- Afghanistan is a consistent low-HDI outlier due to political instability and limited access to education and healthcare.
- Life Expectancy shows a strong positive correlation with HDI, while Gender Development shows a weaker positive relationship.
- Some countries show a mismatch between income and development:
 - Maldives has high income but relatively lower HDI.
 - Nepal and Bangladesh achieve relatively high HDI despite lower income.

Graphical Presentation of Results

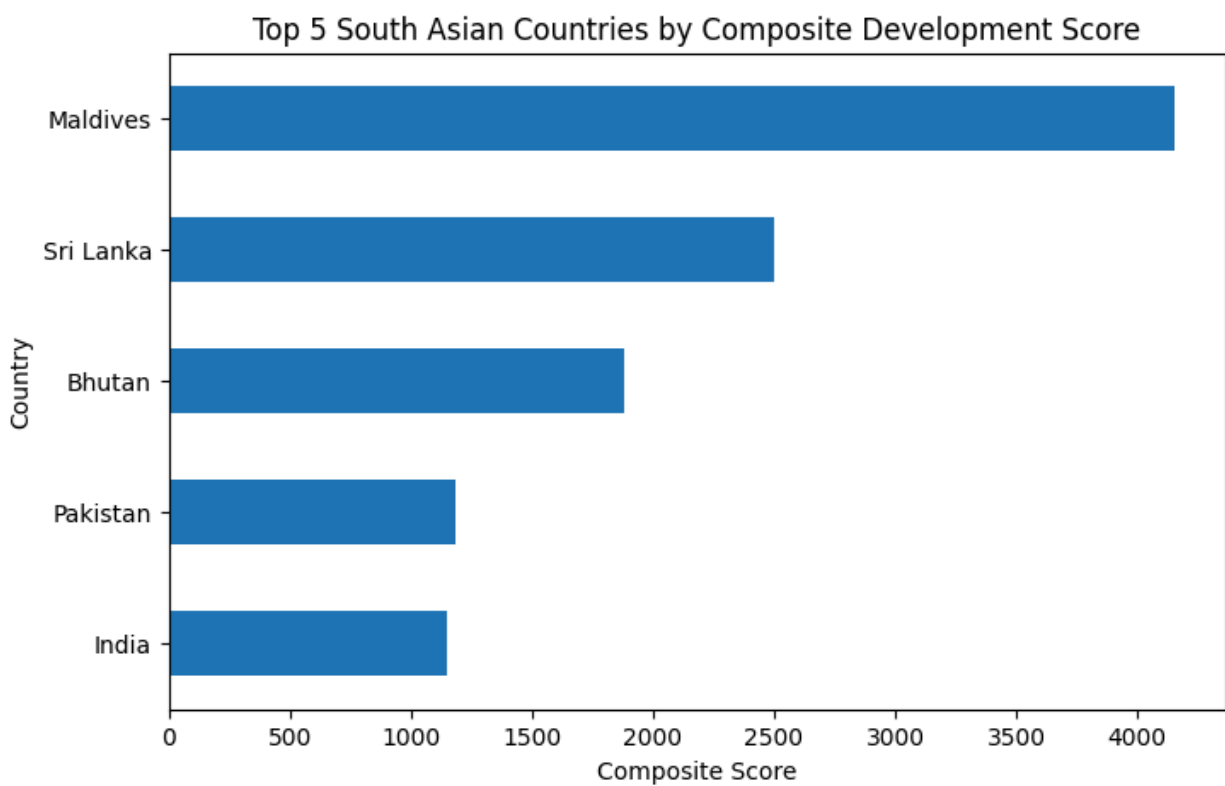
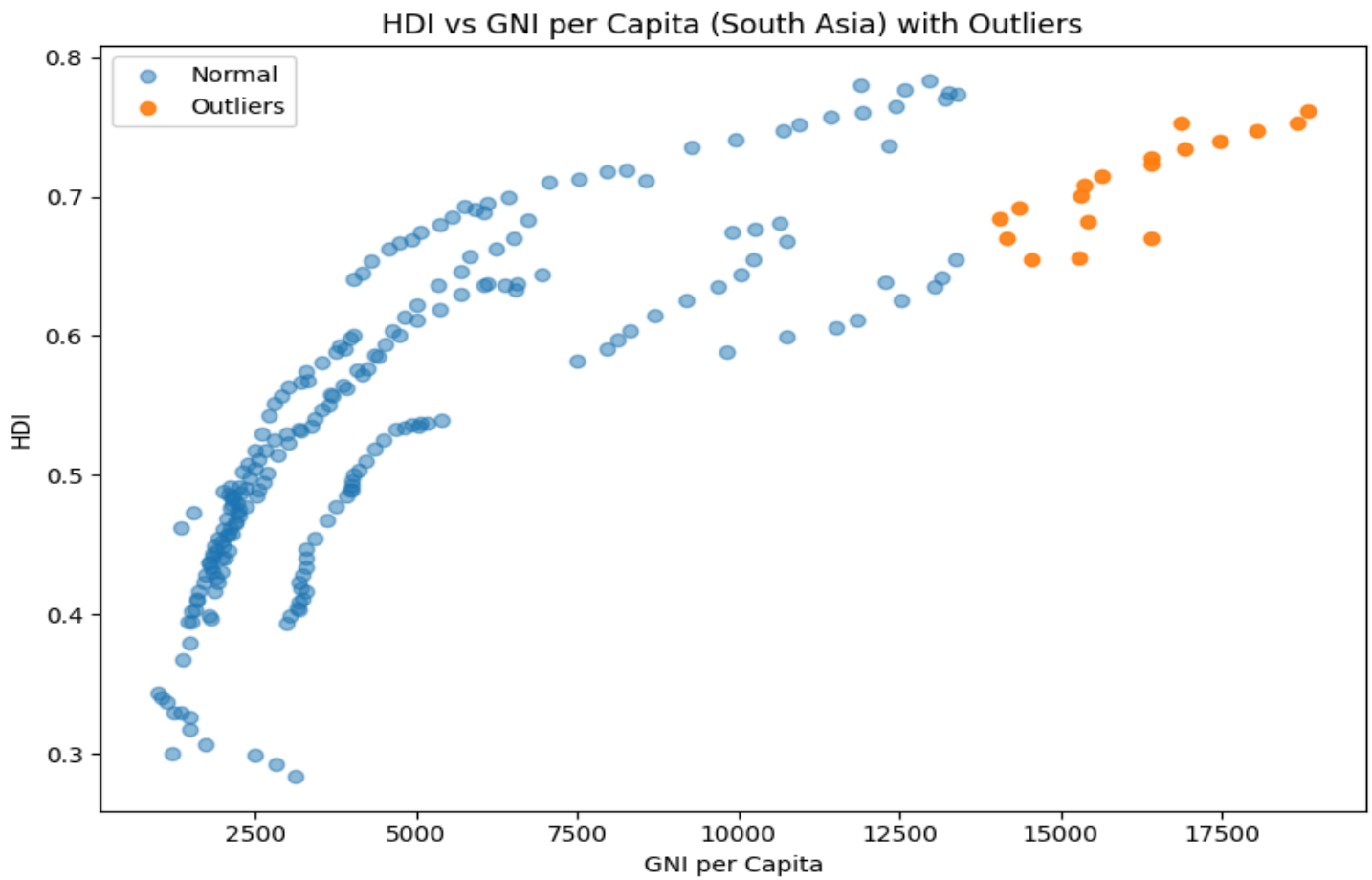


Figure 1: Top 5 South Asian Countries by Composite Development Score



GNI-HDI Gap Extremes in South Asia

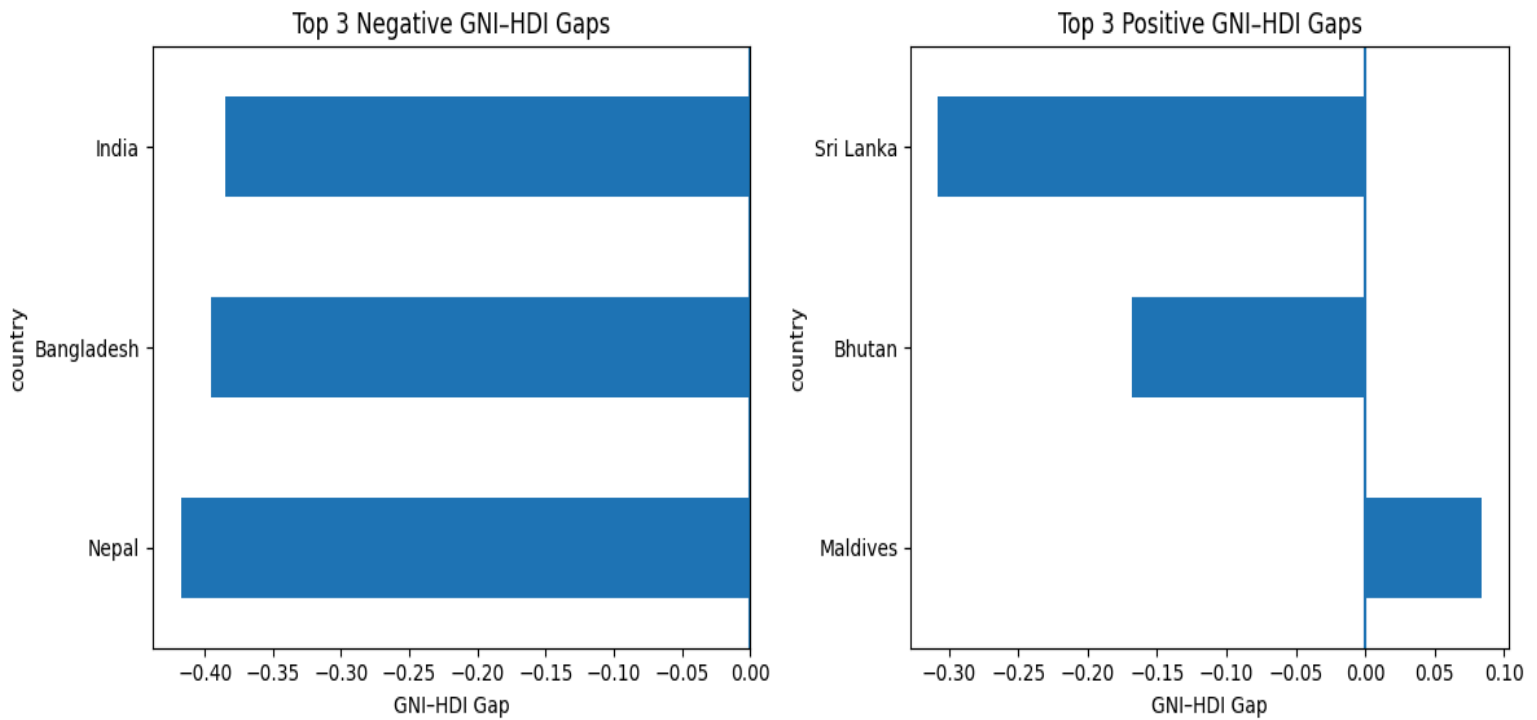


Figure 4: Top Positive and Negative GNI-HDI Gaps

Analysis and Interpretation of Results

The composite development ranking shows that income and health jointly shape development outcomes, sometimes altering country rankings compared to HDI alone.

Life expectancy is the strongest predictor of HDI, confirming that health improvements play a central role in human development. Gender development contributes positively but indirectly, and its effects vary across social and cultural contexts.

The gap analysis highlights that higher income does not automatically translate into better development outcomes. Countries like Maldives show high income but lower-than-expected HDI, suggesting inequality or uneven distribution of resources. Conversely, Nepal and Bangladesh perform relatively well in HDI despite lower income, indicating effective social investments and development policies.

Overall, the findings emphasize that economic growth must be accompanied by inclusive social policies to improve human development.

Problem 3 – Comparative Regional Analysis: South Asia vs Middle East

Methodological Framework

The analysis uses HDI data from 2020–2022 for two regions: South Asia and the Middle East. Countries were grouped into regional subsets, and descriptive statistics were computed to compare overall development levels.

Average HDI, life expectancy, income per capita, and gender development were compared across regions. Top and bottom performers were identified based on average HDI. Regional disparities were assessed using the range and coefficient of variation. Pearson correlation analysis was used to examine relationships between HDI and related indicators. Outliers were detected using the interquartile range (IQR) method.

Main Findings

- The Middle East has a higher average HDI than South Asia.
- The Middle East shows greater inequality in development, with higher HDI range and coefficient of variation.
- Life expectancy and gender development are strongly correlated with HDI in both regions.
- Some Middle Eastern countries show unusually low HDI relative to income, indicating internal disparities.

Graphical Presentation of Results

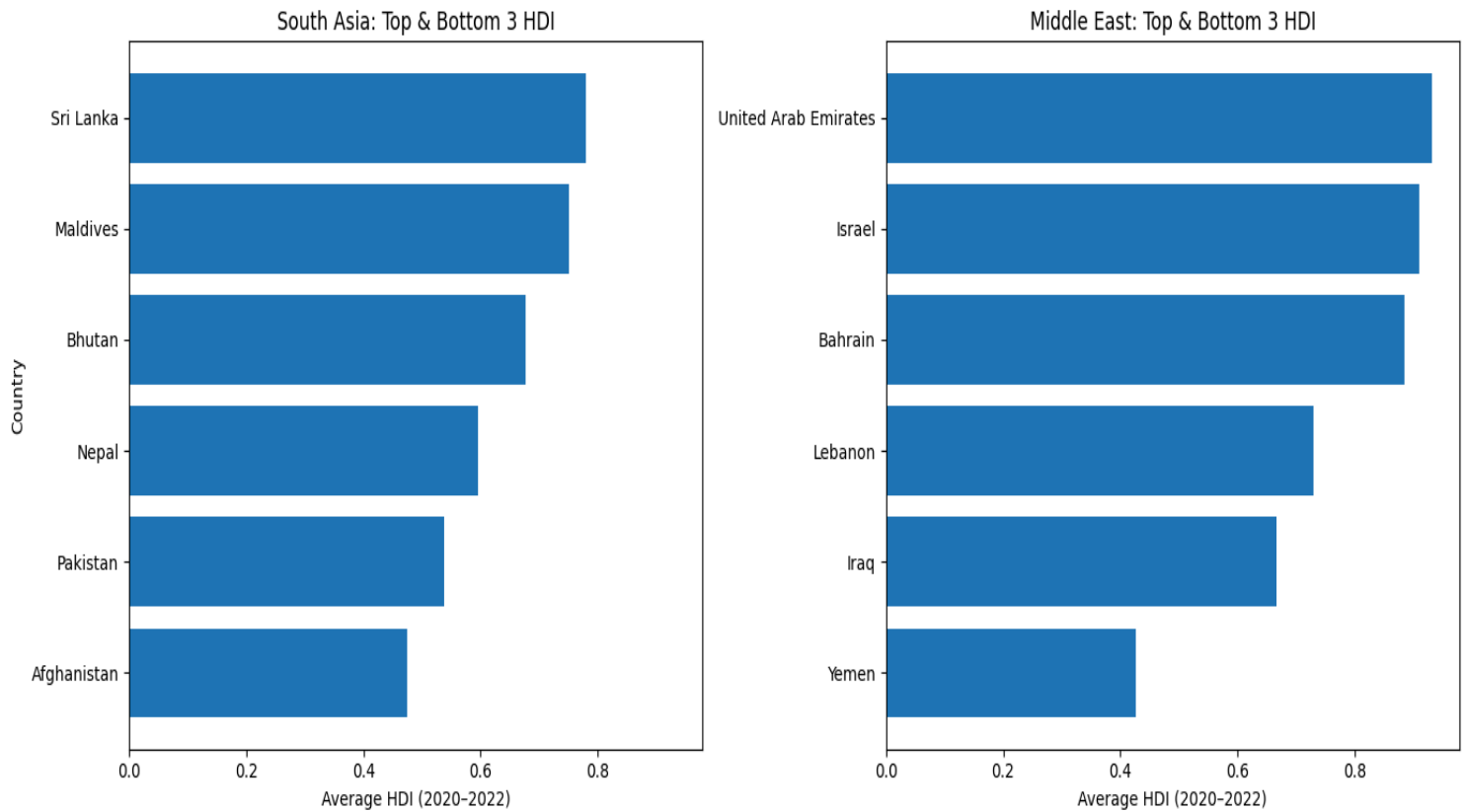


Figure 1: Top and Bottom 3 HDI Countries (South Asia vs Middle East)

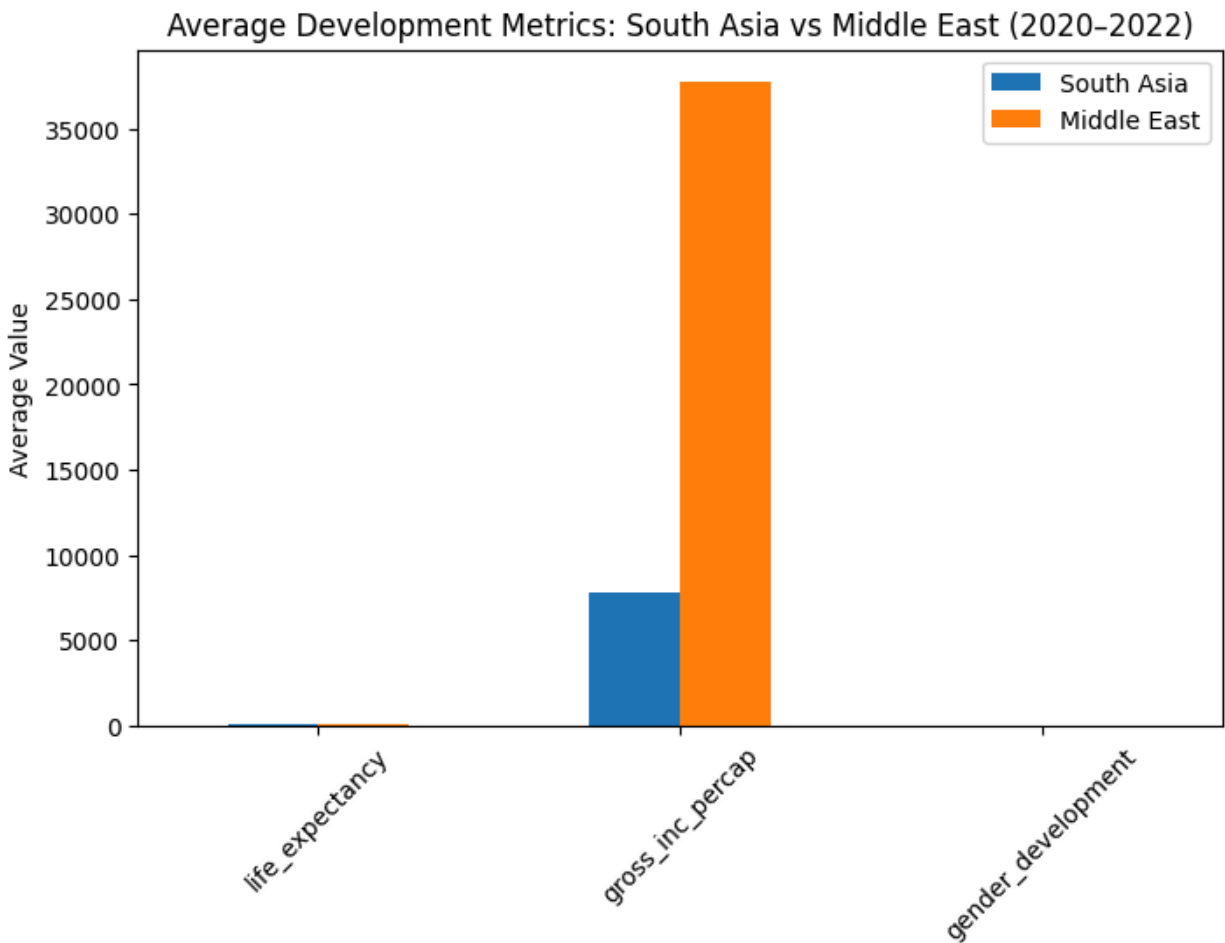


Figure 2: Average Development Metrics by Region

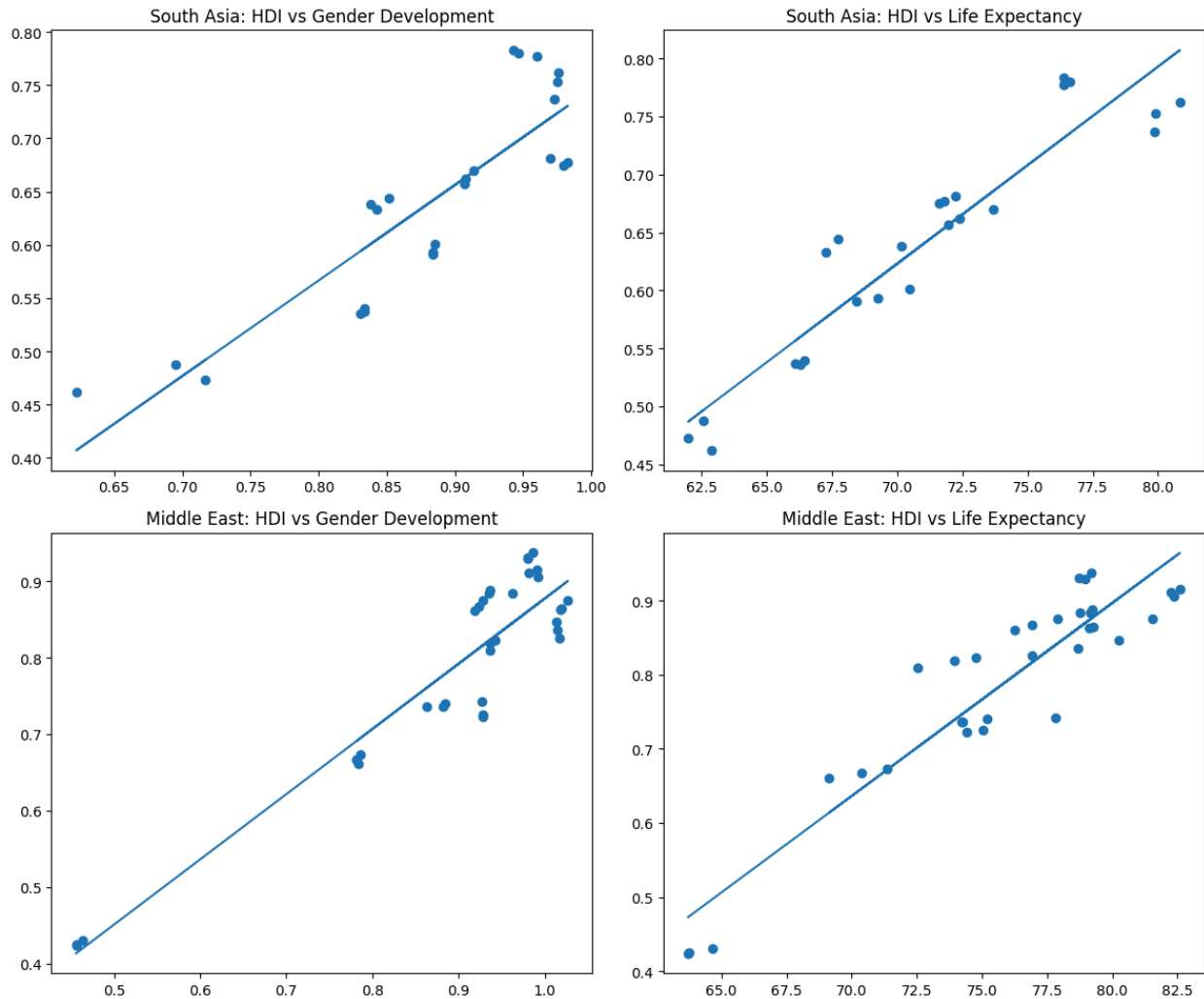


Figure 3: HDI vs Life Expectancy and Gender Development (Scatter Plots)

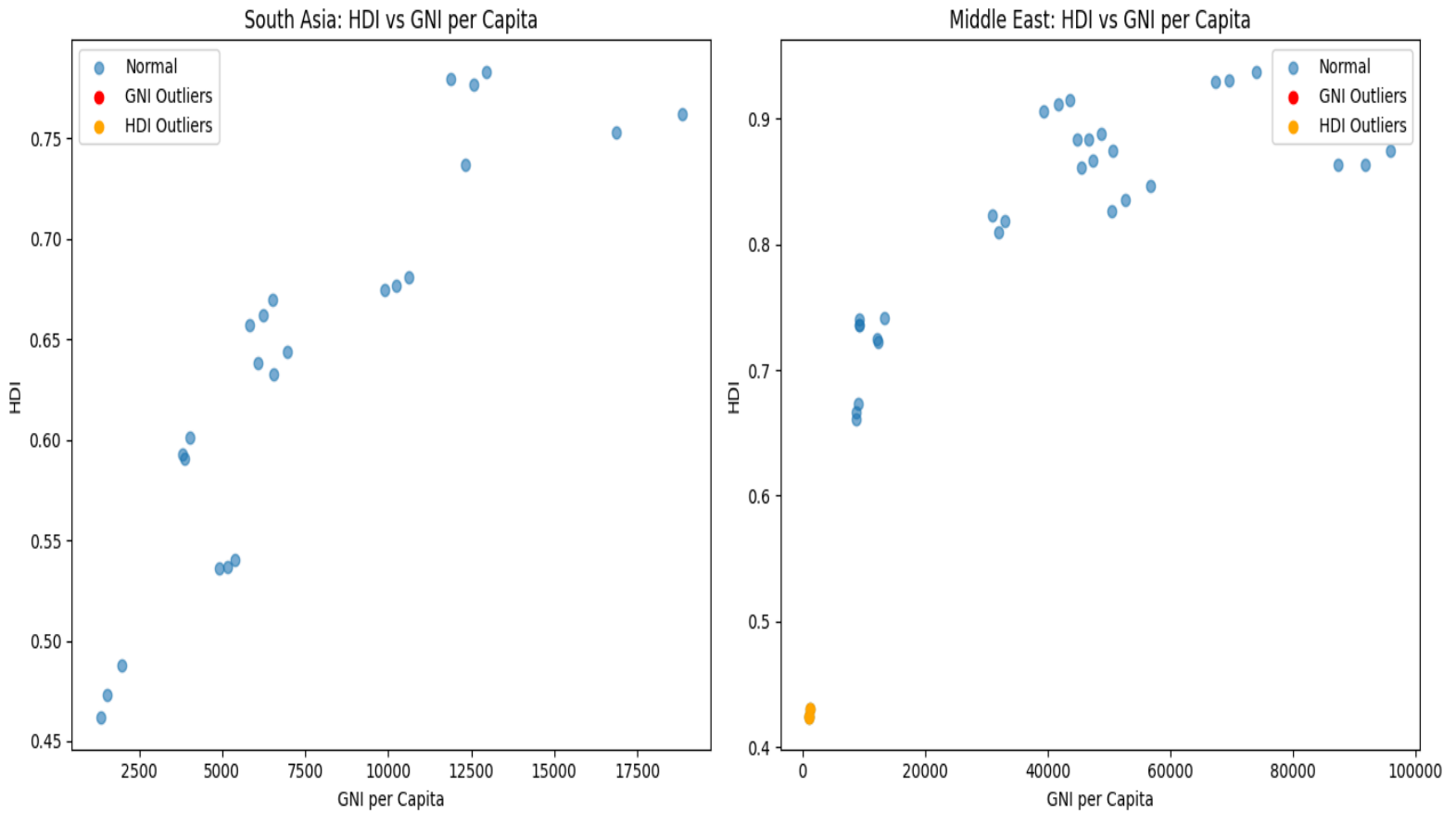


Figure 4: HDI vs GNI per Capita with Outliers Highlighted

Analysis and Interpretation of Results

Figure 1 compares the top and bottom HDI performers in each region. The Middle East contains both very high and very low HDI countries, while South Asia shows a narrower spread. This reflects greater inequality in development outcomes within the Middle East.

Figure 2 shows that the Middle East outperforms South Asia on average in life expectancy, income, and gender development. However, this higher average masks significant internal variation.

Figure 3 shows strong positive linear relationships between HDI and both life expectancy and gender development in both regions. In South Asia, life expectancy shows a slightly stronger association with HDI, while in the Middle East both factors contribute similarly.

Figure 4 highlights outliers with unusually low HDI relative to income in the Middle East, likely due to conflict, political instability, or humanitarian crises.

These outliers increase regional inequality and demonstrate that economic resources alone do not guarantee high human development.

Overall, the analysis shows that although the Middle East has higher average development, it also experiences greater inequality. South Asia has lower average HDI but more consistent development levels. Both regions confirm that health, gender equality, and institutional stability are critical drivers of human development.

5. Concluding Remarks

Overview of Key Outcomes

This study analyzed human development patterns using HDI data across time and regions. The trend analysis showed gradual improvements in HDI for most countries, while the advanced exploration revealed that health and income strongly influence development outcomes. The comparative regional analysis showed that the Middle East has a higher average HDI than South Asia but also exhibits greater inequality within the region.

Insights about HDI Trends and Disparities

Life expectancy consistently emerged as the strongest contributor to HDI variation, confirming the central role of health in development. Gender development also showed a strong positive relationship with HDI, particularly in the Middle East. However, higher income does not always translate into higher human development, as shown by income–HDI gaps and regional outliers. The Middle East displays wider disparities between countries, while South Asia shows lower overall HDI but more uniform development levels.

Limitations

This analysis is limited by data availability and quality, especially for conflict-affected countries where indicators may be incomplete or unreliable. The study period for regional comparison (2020–2022) is relatively short and may be influenced by temporary shocks such as the COVID-19 pandemic. Additionally, HDI does not capture all dimensions of development, such as inequality, political stability, or environmental sustainability.

Practical Implications and Recommendations

Policymakers should focus not only on economic growth but also on investments in health, education, and gender equality to improve development outcomes. Countries with high income but lower HDI should prioritize inclusive social policies to better convert economic resources into human well-being. Future studies could extend this analysis by incorporating inequality-adjusted HDI, environmental indicators, or longer time periods to gain a more comprehensive understanding of development dynamics.